

BCD Model Detailed

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1 Model

Start with the armature equation

$$R_a i_a + L_a \frac{di_a}{dt} + V_b = V_a. \quad (1)$$

Assume negligible armature inductance, so $L_a \approx 0$. For a linear actuator, $V_b = K'_b \dot{z}$ and let $V_a = u$. Then

$$R_a i_a + K'_b \dot{z} = u. \quad (3)$$

Solving for current gives

$$i_a = \frac{u - K'_b \dot{z}}{R_a}.$$

For the basic mechanical model,

$$m_l \ddot{z} = -f_l \dot{z} + K_f i_a.$$

Substitute the expression for i_a :

$$\begin{aligned} m_l \ddot{z} &= -f_l \dot{z} + K_f \left(\frac{u - K'_b \dot{z}}{R_a} \right) \\ R_a m_l \ddot{z} &= -R_a f_l \dot{z} + K_f u - K_f K'_b \dot{z} \\ &= -(R_a f_l + K_f K'_b) \dot{z} + K_f u. \end{aligned}$$

Therefore,

$$\ddot{z} = -\frac{R_a f_l + K_f K'_b}{m_l R_a} \dot{z} + \frac{K_f}{m_l R_a} u. \quad (4)$$

Now define

$$\tau = \frac{m_l R_a}{R_a f_l + K_f K'_b}, \quad K_l = \frac{K_f}{R_a f_l + K_f K'_b}. \quad (5)$$

Using these definitions, the reduced model becomes

$$\tau \ddot{z} + \dot{z} = K_l u,$$

or equivalently,

$$\ddot{z} = -\frac{1}{\tau}\dot{z} + \frac{K_l}{\tau}u. \quad (6)$$

For the extended mechanical model, include the pressure terms:

$$m_l\ddot{z} = -f_l\dot{z} + K_f i_a + \pi r^2 \rho_w g x - \frac{(\pi r^2)^2 P_s}{V_{se}} z.$$

Again substitute $i_a = (u - K'_b \dot{z})/R_a$:

$$\begin{aligned} m_l\ddot{z} &= -f_l\dot{z} + K_f \left(\frac{u - K'_b \dot{z}}{R_a} \right) + \pi r^2 \rho_w g x - \frac{(\pi r^2)^2 P_s}{V_{se}} z \\ R_a m_l \ddot{z} &= -R_a f_l \dot{z} + K_f u - K_f K'_b \dot{z} + R_a \pi r^2 \rho_w g x - \frac{R_a (\pi r^2)^2 P_s}{V_{se}} z \\ &= - (R_a f_l + K_f K'_b) \dot{z} + K_f u + R_a \pi r^2 \rho_w g x - \frac{R_a (\pi r^2)^2 P_s}{V_{se}} z. \end{aligned}$$

Divide by $R_a f_l + K_f K'_b$, and use

$$\frac{\tau}{m_l} = \frac{R_a}{R_a f_l + K_f K'_b}.$$

Then

$$\tau \ddot{z} + \dot{z} = K_l u + \left(\frac{\tau}{m_l} \right) \pi r^2 \rho_w g x - \left(\frac{\tau}{m_l} \right) \frac{(\pi r^2)^2 P_s}{V_{se}} z.$$

Finally,

$$\ddot{z} = -\frac{1}{\tau}\dot{z} + \frac{K_l}{\tau}u + \frac{\pi r^2 \rho_w g}{m_l}x - \frac{(\pi r^2)^2 P_s}{m_l V_{se}}z. \quad (7)$$